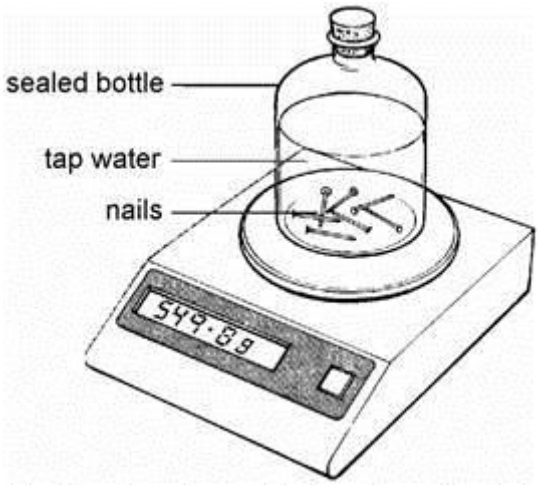
Q1.

An experiment was set up to investigate rusting. Some clean, shiny, iron nails were sealed in a glass bottle containing some tap water. The sealed bottle was then placed on a top-pan balance. The reading on the balance was 549.8 g.



The sealed bottle was left for one week. After one week the nails were rusty.

(a) (i) What would you expect the reading on the balance to be after one week?

.....

1 mark

(ii) Give a reason for your answer.

.....

1 mark

(b) (i) Rust is an oxide of iron. Another oxide of iron is iron(III) oxide.
Write a word equation for the formation of iron(III) oxide from its elements.

.....

1 mark

(ii) Which one of the following words describes the formation of iron(III) oxide from its elements?

combustion **condensation** **decomposition** **oxidation**

.....

1 mark
Maximum 4 marks

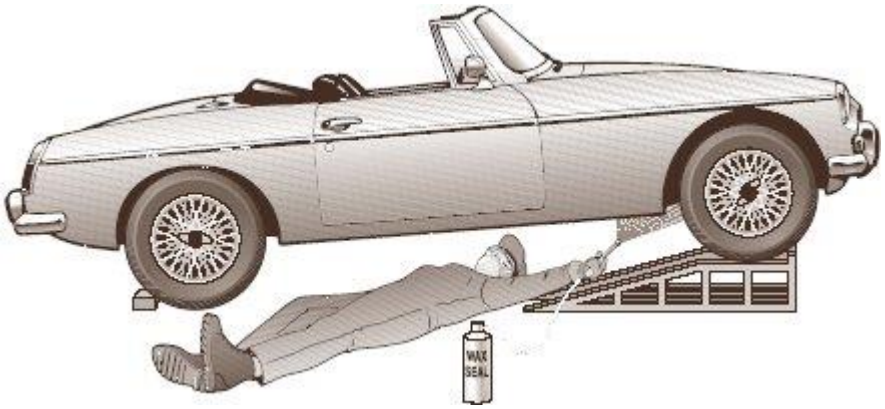
Q2.

Jill bought a can of Wax Seal to spray the parts underneath her car.



Wax Seal helps to prevent these parts rusting.

It is a mixture of wax and a liquid called white spirit.



(a) (i) The body of Jill's car is made from steel. Steel contains iron.

Give **two** substances that are needed for iron to rust.

1. .....

1 mark

2. .....

1 mark

(ii) How does Wax Seal help to protect the car from rusting?

.....

.....

1 mark

(iii) Wax Seal can also be used on the upper parts of a car.

What else protects parts such as the doors from rusting?

.....

1 mark

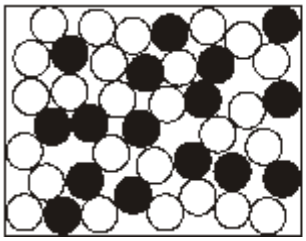
(b) The metal parts of a car may corrode.

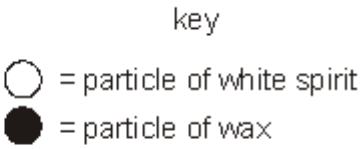
What type of air pollution could cause corrosion?

.....

1 mark

(c) The diagram below shows the mixture of particles of wax and white spirit in Wax Seal.





not to scale

After Jill sprays the car, the white spirit evaporates leaving a layer of solid wax on the surface.

(i) In the box below, draw **eight** circles, ○, to show the arrangement of particles in a gas.

particles in a **gas**

1 mark

(ii) In the box below, draw **eight** circles, ●, to show the arrangement of particles in a solid.

particles in a **solid**

1 mark
maximum 7 marks

Q1.

(a) (i) 549.8 g

*accept 'the same' **or** 'no change'*

1 (L5)

(ii) mass does not change in a reaction

accept 'because it is sealed'
***or** 'new material is not made **or** lost'*

*do **not** accept 'mass does not change'*

1 (L5)

(b) (i) iron + oxygen → iron(III) oxide

the whole word equation is required for the mark

accept 'iron oxide' for iron(III) oxide reactants
may be in either order

*do **not** accept 'air' for oxygen*

1 (L6)

(ii) oxidation

1 (L6)

[4]

Q2.

(a) (i)

- oxygen

1 (L6)

- water

1 (L6)

answers may be in either order

'air' is insufficient

*'moisture' **or** 'dampness' **or** 'wet' are insufficient*

(ii) any **one** from

- it prevents contact between the steel **or** the car and oxygen **or** water
- it is waterproof **or** water runs off

accept 'it prevents air getting to the car'

*accept 'wax fills scratches **or** chips where paint is damaged'*

'it forms a protective layer' is insufficient

1 (L5)

(iii) any **one** from

- paint
- chrome

accept 'they are coated in zinc'
or 'they are galvanised'

accept 'polish'
'rust treatment' is insufficient
'cover it' is insufficient

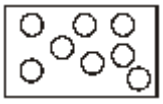
1 (L5)

(b) • acid rain

accept 'sulphur dioxide'
accept 'oxides of nitrogen'
accept 'car exhaust fumes'
accept 'burning fossil fuels'
accept 'sea air' ***or*** *'salty air'* ***or*** *'salt'*
'carbon dioxide' is insufficient

1 (L6)

(c) (i) • *gas*: particles randomly arranged and most **not** touching



accept black shaded circles if drawn correctly
accept fewer ***or*** *more than 8 circles if the arrangement is clear*
ignore arrows attached to circle

1 (L6)

(ii) • *solid*: particles regularly arranged and all touching



accept white circles if drawn correctly
accept 2 rows of particles with at least 2 particles in the second row
accept fewer ***or*** *more than 8 circles if a regular arrangement is clear*
ignore location of circles in box
do ***not*** *accept a single row of circles*

1 (L6)

[7]